Mini-Project Report
Revising Methods and Performing Evaluation

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Abstract

Modern data visualization dashboards are widely used to support analysis and decision-making, but design choices and interaction techniques shape how users interpret the data they present. This mini-project examined whether adding interactive features such as data exploration, filtering, and line isolation improves a user's ability to interpret time-series data compared to a static visualization presenting the same information. Two versions of a dashboard were built in Tableau Public using per-capita energy consumption data from *Our World in Data*, differing only in their level of interactivity. Six participants completed six interpretation tasks across the static (n=2) and interactive (n=4) visualizations, with performance measured through task accuracy, completion time, Likert-scale ratings of clarity and confidence, and open-ended written responses. Interactive participants reported substantially lower cognitive load and higher ease of use, while accuracy was comparable across conditions and mean completion time was longer in the interactive condition. Half of participants independently requested explicit indicators of year over year change, a feature neither version provided. These findings will inform a continuation of the project in senior comps, focused on directional change indicators and a potentially methodologically stronger within-subjects evaluation.

1 Introduction

Modern data visualization dashboards are becoming increasingly popular and important across industries such as healthcare, business, and manufacturing to support data analysis, decision-making, and communication [21, 7, 1]. However, prior research has shown that visualizations are not neutral representations of data [6, 16]; design choices and interaction techniques can significantly shape how users interpret information [5, 15]. To reduce misinterpretation and promote a shared understanding of data visualizations/dashboards, a multifaceted approach is needed that takes into account the human variable with interactive design, because misinterpretation doesn't stem from user error but from a combination of design choices, interaction techniques, and human differences [20, 23, 10, 4, 12, 19].

This mini-project will thus focus on the role that interactivity plays in dashboard interpretability. More specifically, it will examine whether adding interactive features, such as filtering and user driven exploration, found on many of the most widely used dashboard applications will improve a user's ability to correctly interpret data compared to a static visualization presenting the same information. By implementing two versions of a dashboard that differ only in their level of interactivity, the project will isolate the effect of interaction on interpretation accuracy, perceived clarity, and confidence.

Within the four week project period, I designed a functional albeit simple interactive and static dashboard prototype, defined specific interpretation tasks based on the primary energy consumption per capita dataset that I gathered from *Our World in Data* [22], and conducted a small scale user study. This work is relevant to broader areas of computer science, including data analytics, human computer interaction, and information visualization, and aligns with my interest in understanding how researchers and data analysts are able to promote analytical reasoning through their visualizations and interactive dashboards.

2 Methods

2.1 Citing Sources for your Methods

The methods chosen for this mini-project are grounded in established practices in data visualization and human-computer interaction research. Prior literature using task-based evaluation methodologies, where participants completed predefined analytical tasks using a system, has been used to provide an objective measure of how effectively a system supports user understanding and decision-making [13, 24].

The six tasks designed for this study were modeled after low level analytical activities [2, 25], such as retrieving values, finding extrema, and comparing trends across categories, which are among the most common and well validated task types in visualization evaluation. Additionally, open-ended questions were included to gather insights into user reasoning that may not have been fully observable through predefined tasks or quantitative metrics alone [18, 14, 3].

One method I initially considered but ultimately did not pursue was a thinking aloud method, where participants are told to verbalize their reasoning while interacting with the system. While this approach has been used in prior visualization studies to evaluate cognitive processes as they occur, I believed it to be impractical within the four week time constraint and small participant pool. Instead, post-task open ended written responses were used as a more feasible substitute that still captures qualitative reasoning.

2.2 Justifying Your Methods

A controlled comparison between static and interactive dashboard versions which use the same data and design choices remained the most appropriate method for isolating the effect of interactivity on interpretability, as prior research argued that the effects of interaction cannot be meaningfully assessed without controlling for other design variables [25]. Both versions included a bar chart for snapshot comparisons and a line graph for temporal trends so any differences in user performance could be attributed specifically to the presence or absence of interactive features such as filtering and country selection.

Task based user studies are among the most widely used and appropriate methods for assessing visualization interpretability [17, 13, 24], and the tasks here were deliberately designed to span a range of reasoning demands: from precise value retrieval (Q1, Q6) to trend comparison and open-ended pattern recognition (Q4, Q5). This was done intentionally because by only relying on accuracy based tasks, I would potentially be overstating interpretability by only capturing performance on the lowest analytical level.

2.3 Materials Needed for Your Methods

The dataset used is a publicly available energy consumption per capita dataset from *Our World in Data*, which aggregates data from the U.S. Energy Information Administration and the Energy Institute's Statistical Review of World Energy, covering all countries over several decades. This dataset was selected because it contains both temporal and categorical variables that benefit from filtering and trend comparison tasks and because I also found the comparisons to be non trivial and analytically interesting.

Both dashboard versions, static and interactive, were built in Tableau Public, the free version of the Tableau data visualization platform. Tableau was chosen over alternatives such as PowerBI or other Python-based tools (e.g., Plotly, Dash) because it natively supports both fully static exports and interactive dashboards within the same environment, making it possible to produce comparable versions that differ only in interactivity level.

A Google Form was created to consolidate the evaluation materials, which collected brief background information upon the 6 participants such as their familiarity with data visualizations to help contextualize any variation in performance across the sample. It also included six interpretation questions, a short Likert scale questionnaire measuring perceived clarity and confidence, and a few open-ended follow-up prompts.

3 Schedule

The mini-project was completed over four weeks, with each week focusing on a distinct phase of planning, selecting, implementing, and evaluating. I've noted the week-by-week progression below and on Table 1.

Week 1: The first week focused on reviewing additional targeted literature that specifically relates to interaction and interpretability in data visualization. This review informed specific design decisions for the dashboard and helped refine evaluation metrics and interpretation tasks.

Week 2: During the second week, I searched for and selected an appropriate dataset and finalized the dashboard design. This included deciding on the specific visual encodings, layout, and interaction techniques that were implemented the following week.

Week 3: The third week was dedicated to the creation of the dashboard prototypes (both static and interactive), the development of the interpretation tasks, and the preparation of user evaluation materials (Likert scale/open-ended questions).

Week 4: In the final week, I recruited participants of varying backgrounds, conducted the user study, and evaluated the results. Data collected from the study was analyzed, and the project concluded with this reflection on findings, limitations, and potential directions for future work.

3.1 Challenges of Scheduling

The mini-project largely followed the original four-week plan, with each week corresponding to a distinct phase of the project. The primary challenge was time management in Week 3, where the dashboard, task design, and evaluation material all had to be completed in parallel, which stretched into Week 4. This may be why I felt a bit time constraint during Week 4 with the user study portion as I originally planned to test my dashboards with more users.

4 Evaluation

Evaluation was focused on whether dashboard interactivity improves accurate and timely interpretability and

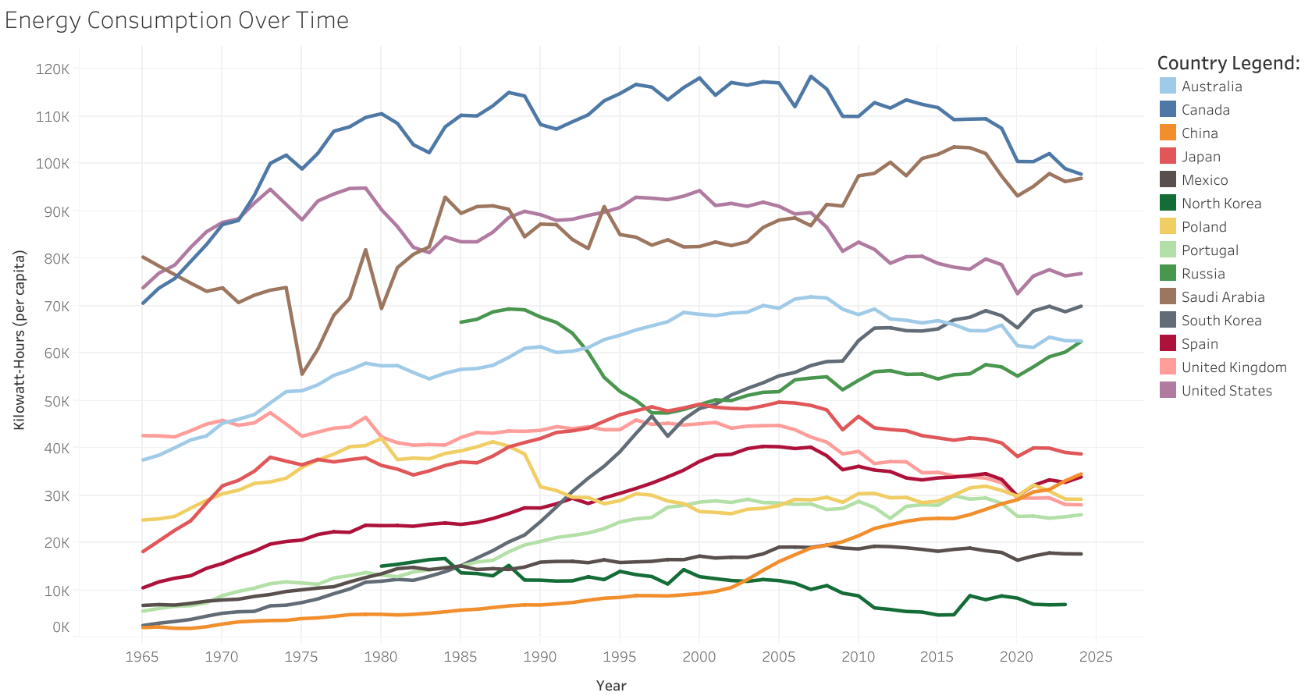

Figure 1: Interactive users were able to highlight a country by clicking on its name/line which would make them stand/pop out against the other countries, and were able to filter them out by clicking on an "Exclude" button not shown here.

Week	Goal	Stretch Goal
Week 1	Reviewed dedicated literature	Started the dataset search
Week 2	Finalized dataset search and dashboard design	Familiarization with Tableau
Week 3	Dashboard, task, and question creation	Did not begin user studies
Week 4	Conducted user studies, analyzed data, and reflection	Light qualitative analysis

Table 1: Actual progression of the mini-project

User	Version	Time
1	Interactive	8 minutes 50 seconds
2	Interactive	12 minutes 35 seconds
3	Interactive	4 minutes 25 seconds
4	Interactive	8 minutes 50 seconds
5	Static	7 minutes 35 seconds
6	Static	5 minutes 40 seconds

Table 2: Total time each user took to complete all 6 tasks

whether it was able to reduce misinterpretation when compared to a static visualization. Quantitative and qualitative measures were both be used here to capture the nuances of human understanding.

4.1 Metrics

I implemented task completion time as an added metric beyond what was initially planned for with task accuracy as findings suggest that efficiency is a meaningful dimension of visualization effectiveness alongside correctness since users may answer correctly while still experiencing high cognitive strain, making time a useful proxy for interaction fluency [17, 8].

A Likert scale and open-ended questions were also included to capture subjective aspects of user understanding like perceived clarity and confidence that are not directly observable through task performance alone which led to a better understanding of how effectively a visualization supports understanding by providing some qualitative insights into a participant's reasoning processes and potential sources of misinterpretation [17, 11, 14, 9].

5 Results

Six participants with varying backgrounds completed the questionnaire via a single Google Form, with two being assigned to the static visualization and four to the interactive one. Two users reported prior experience with tools such as Tableau or Power BI (one in each condition), and self rated visualization familiarity averaged 3.5 on a 1–5 scale, which was comparable between static and interactive version. The results below are reported across the four metric categories: task accuracy, task completion time, self-reported clarity

Item	Static (n=2)	Interactive (n=4)
Easy to identify trends	5.00	5.75
Easy to compare countries	5.00	5.00
Visualizations easy to use	5.00	6.50
Felt overwhelming	5.50	2.75
Difficult to track multiple countries	5.50	3.00
Confident in answers	6.50	5.75
Interactive features helped	——	6.50
Filtering helped focus	——	6.75

Table 3: Self-reported responses using Likert means (1–7)

and confidence, and qualitative open-ended responses.

The six interpretation tasks varied between value retrieval (Q1: identifying the highest-consuming country in 1971), trend judgment (Q2: identifying the country with the most stable trend; Q3: identifying the country with the largest increase over time), open-ended trend comparison between two user-selected country pairs (Q4: countries with differing trends; Q5: countries with similar trends), and rank comparison across adjacent years (Q6: whether country rankings changed between 1993 and 1994).

Task Accuracy: Users generally performed well across both conditions, though accuracy was notably higher on tasks requiring precise value retrieval and trend reasoning (Q1, Q2, Q3) than on tasks requiring rank comparison (Q6), as shown in Table 4. The bar chart and year filter combination was pretty useful in Q1 and Q6, where users needed to identify or compare specific values at a given point in time.

Task Completion Time: Users who were presented with the static version had lower average times (6 minutes 37 seconds) vs interactive users (8 minutes 40 seconds), with the highest individual time of 12 minutes and 35 seconds coming from an interactive participant who reported ”constantly clicking” and ”taking my time” to isolate lines, as shown in Table 2.

Likert Scale Responses: Self-reported clarity and confidence scores were consistently higher in the interactive condition across most tasks, as shown in Table 3. Users rated the interactive dashboard as clearer, especially for tasks involving multiple countries over time (Q4, Q5), where overlapping lines in the static version added visual clutter.

Open-Ended Responses: Through the qualitative responses, I figured out that the interactive users were more likely to reference specific filtering actions as part of their reasoning process; for example, noting that they narrowed the view to two or three countries before drawing conclusions. Static users were more frequently expressed uncertainty, particularly on Q2 and Q5, where identifying stable or similar trends required mentally parsing a dense, unfiltered line graph, as shown in Figure 1.

6 Discussion

The clearest pattern as reported in Table 3 is a reduction in self reported cognitive load with the interactive visualization since ratings on the ”visualization felt overwhelming” and ”it was difficult to track multiple countries” questions fell by 2.75 and 2.50 points on a 7-point scale respectively, with ease-of-use rising by 1.50 points. These differences are pretty large despite the small sample size which aligns with what the participants shared: the static users described the difficulty of tracking individual countries through overlapping line segments, while interactive users described filtering and isolation as the techniques that made their reasoning controllable. This suggests that the dashboard's interactive features are meeting their purpose, not by changing what answers participants reach, but by changing how effortful those answers feel to produce.

However, I do have to note that two results complicate the simple “interactive is better” statement. First, mean completion time was longer in the interactive condition (8:40 vs. 6:37), opposite to my pre-study expectation that filtering would shorten comparison tasks. The best explanation I can provide is that interactive features invite and promote user exploration, as the behavior of the user with longest individual time may also be what produces the lower cognitive load ratings that they reported. Another argument is that the interactive interface introduces interaction overhead, which is time spent toggling, filtering, and re-filtering, that exceeds the time it saves on the comparison itself. Second, self-reported confidence was marginally higher in the static condition (6.50 vs. 5.75), and the only explanation I can articulate is that static users had fewer ways to second guess their initial reading and so they reported higher subjective certainty.

It is also worth noting the limitations that plague these findings. The biggest of them being the small participant pool of six, especially the number of static users only being two, which limits the generalization of these findings. The sample is also demographically narrow (ages 17–21, predominantly college students, with computer science overrepresented), which also limits generalizability to broader audiences. Time and recruitment constraints did not help as well with the scale of the user study. Completion times were self-reported rather than logged by the dashboard, so there is probably some error in the seconds with the recorded times. If this project is expanded upon for the senior comps, work should be done to recruit a larger and more diverse sample, see if it's possible to automatically log the time and interactions on the dashboard, and think about a within subjects design where users utilize both interactive and static versions to better account for the human variability of this study, as was done in [6].

Table 4: Per-question accuracy rates by condition. Q1–Q3 required identifying a single country; Q6 required identifying whether (and how) the country ranking changed between 1993 and 1994. Q4 and Q5 were open-ended and not scored for correctness.

Question	Task type	Static (n=2)	Interactive (n=4)
Q1: Highest consumer in 1971	Value retrieval	2/2 (100%)	4/4 (100%)
Q2: Most stable trend	Trend judgment	2/2 (100%)	4/4 (100%)
Q3: Largest increase over time	Trend judgment	2/2 (100%)	4/4 (100%)
Q6: Rank change 1993 vs. 1994	Rank comparison	1/2 (50%)	2/4 (50%)*

*One additional interactive participant gave a partially correct answer naming countries whose values changed without explicitly identifying a rank swap.

7 Conclusion

During this four-week long mini-project, I successfully managed to build and pilot test an interactive dashboard for exploring per-capita energy consumption across countries from 1965 to the present, which was evaluated against a static counterpart through a six participant questionnaire study. Results showed that interactivity substantially lowered cognitive load and higher ease of use, crediting filtering, line highlighting/isolation, and sorting as the features that made reasoning manageable. At the same time, the results do not tell a uniformly positive story: mean completion time was longer in the interactive version, self-reported confidence was marginally higher in the static version, and accuracy was comparable across both. The most consistent advice for improvement across both versions came about qualitatively through the open-ended responses as half the participants independently asked for explicit numeric or directional indicators of change, a feature neither version currently provides.

As I currently stand, I am planning on continuing this work into senior comps. I am definitely planning on addressing the concerns and add directional indicators for year over year change with optional numeric annotations. I will also work to create a methodologically stronger evaluation by exploring a within subjects study with a larger and more demographically diverse participant pool and scoring of free response trend comparisons through a predefined rubric. Together, these initial improvements should move my project from being a working prototype to a credible contribution that supports research on how interactivity shapes interpretation in data visualizations.

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